

Juan David Vela Coronado

Estudiante de Ingeniería de Sistemas — 9º semestre • Universidad del Cauca

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Systems Engineering student (9th semester) with hands-on experience building **distributed, microservices-based software** in teams that deliver to **real clients**. I have **led a 5-developer team under Scrum**, coordinated remote work through Git-based workflows and code review, and shipped a platform to production for a public-sector client. Comfortable across **Java (Spring Boot)**, **Node.js (NestJS)** and **Python (Flask)**, with cloud deployments on **AWS** and **Docker**.

PROJECT EXPERIENCE

Community Action Boards (JAC) Management Platform

Popayán, Colombia

Team Lead & Backend Developer • Client: Gobernación del Cauca

2026 • Live Deployment

- **Led a 5-developer student team** from Universidad del Cauca building a platform for the **Gobernación del Cauca**: I defined the **technology stack**, negotiated **project scope** with the client, and **assigned tasks according to each member's skill level**.
- Ran the project under **Scrum**: sprint planning, code review, and **delivery demos directly with the client**, gathering and refining requirements each iteration.
- Architected a **microservices platform** with **NestJS (Node.js)** to manage Community Action Boards (JACs) and Asocomunales, covering government user management, organizational data, and legacy data migration.
- Implemented **authentication and authorization with Keycloak** and **Google OAuth (Google Login)** for secure, multi-role access to a production government system.
- Delivered a **data migration module** to streamline updating institutional records; **containerized services with Docker** and shipped with a **React 19** frontend.

AIMO — Generative AI Conversational Assistant

2026

Backend Developer • Distributed team, Git-based workflow

Live Deployment

- Built the backend for a **Generative AI** assistant delivering real-time emotional support to university students, serving empathetic, safe, and contextualized responses based on the user's risk level.
- Designed a **3-agent cascading pipeline with continuous real-time evaluation** using **Python 3.13** and **Flask 3.0**, orchestrating inference across the **Groq API** and **AWS Bedrock** (Amazon Nova Pro).
- Engineered prompt orchestration and safety guardrails to guarantee reliable, context-aware model serving; **containerized with Docker** and deployed to production with a React frontend.

Desafío Strata Analytics — AI Agents Challenge

Popayán, Colombia

Individual participant — Autonomous customer-service agent

6th of 24 evaluated agents • March 2026

- Placed **6th out of 24 evaluated agents** (47 registered participants) in a datathon where each participant designed and shipped an **autonomous customer-service agent** for a retail scenario.
- Implemented **tool calling, retrieval (RAG)** over policy documents and a **DynamoDB-backed data layer on AWS Bedrock**.

Alzheimer's Stage Classification — Machine Learning Pipeline

2025

Machine Learning Engineer

Best Poster Award — Universidad del Cauca (2026)

- Built a **machine learning pipeline** to predict Alzheimer's severity levels from medical imaging using **Convolutional Neural Networks (CNNs)** and **transfer learning**.
- Awarded **Best Poster** at Universidad del Cauca's student project showcase for the project's technical depth and impact.

RESEARCH EXPERIENCE

Data Science Research Seedbed (*Semillero de Investigación en Ciencia de Datos*) Popayán, Colombia

Member • Faculty of Electronics and Telecommunications (FIET), Universidad del Cauca September 2025 – Present (~11 months)

- Active member for over **11 months**, mentored by Prof. Néstor Milciades Díaz Mariño, researcher of the **Computational Intelligence Research Group**, working on data science and machine learning topics alongside my regular coursework.

EDUCATION

Universidad del Cauca

Bachelor's Degree in Systems Engineering — 9th semester

Popayán, Colombia

2022 – Expected Graduation: 2027

Coursework relevant to GDSD (grade / 5.0): Distributed Systems 4.9 • Software Engineering III 4.8 • Software Engineering II 4.7 • Artificial Intelligence 5.0 • Operating Systems 4.6 • Software Quality 4.5 • Research Methodology 4.5 • Enterprise Software Architectures 4.4 • Databases I–II 4.3 • Networks 4.1.

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript/TypeScript, Go.
- **Backend:** Spring Boot, NestJS (Node.js), Flask, REST APIs, Microservices, WebSockets, gRPC.
- **Frontend:** React, Angular, TypeScript, HTML, CSS.
- **AI / ML:** LLM & Agent Pipelines, Deep Learning (CNNs, Transfer Learning), AWS Bedrock, Groq API.
- **Cloud & Security:** AWS, Keycloak, OAuth2 / Google Login, Role-Based Access Control (RBAC).
- **Databases:** PostgreSQL, Oracle Database, MariaDB.
- **Collaboration & Practices:** **Agile (Scrum)** — sprint planning, dailies, retrospectives; **Git/GitHub** — branching, pull requests, code review; Docker, Postman, Linux; technical documentation written in English.

LANGUAGES

- **Spanish:** Native.
- **English:** Upper-Intermediate (B2 or above) in reading, listening, speaking and writing. Working language for my technical documentation, source code and professional profile.

CERTIFICATIONS AND CERTIFIED COURSES

EF SET English Certificate — 60/100, level B2 (Upper Intermediate) — EF SET 2026
Reading: C1 • Listening: C1 • Writing: B2 • Speaking: B2

Getting Started with Deep Learning — NVIDIA 2026
Credential ID: M0S7oiZMQcO9R966P9O6-Q

CCNA: Introducción a las Redes — Cisco Networking Academy 2026
Duración: 70 horas

CCNA: Fundamentos de Conmutación, Enrutamiento y Redes Inalámbricas — Cisco Networking Academy 2026
Duración: 70 horas

Junior Cybersecurity Analyst Career Path — Cisco 2025
Credential ID: a917b782-b2b7-47d4-bf1c-5d370a6df267 • Duración: 70 horas

Full Stack Empresarial con Spring Boot y Angular — DevSenior 2025
Credential ID: e5f3a1e1-5caf-47cb-833e-a6a5b8f3629a

Los certificados correspondientes se anexan como soportes adicionales (ítem 6 de la convocatoria).